



Runtime Scaling Tables

Table 1: Runtime Scaling Tables by Problem Size

Source data: runtime_results.csv.

Rows are aggregated from raw runs using the grouping fields scale, m, n, unknowns, nodes, ppn, threads, mpi_ranks, total_cores, unknowns_per_core, and matrix_type.

$\text{speedup} = \text{baseline runtime} / \text{runtime mean}$,

$\text{efficiency} = \text{speedup} / \text{total cores}$. CUs = nodes × runtime mean / 3600

Problem Size Overview

scale	m	n	unknowns
small	1,000	1,000	1,000,000
medium-small	2,240	2,240	5,017,600
medium	3,160	3,160	9,985,600
large	4,500	4,500	20,250,000
very-large	6,400	6,400	40,960,000

Table 1a. Runtime scaling for small problem size

scale: small
problem size: 1,000 × 1,000
unknowns: 1,000,000

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
1	1	1	1	1	1,000,000	246.85	1.00	100.0%	0.0686
2	1	1	1	2	500,000	230.50	1.07	53.5%	0.0640
2	1	2	2	1	500,000	83.81	2.95	147.3%	0.0233
4	1	1	1	4	250,000	228.10	1.08	27.1%	0.0634
4	1	2	2	2	250,000	78.84	3.13	78.3%	0.0219
4	1	4	4	1	250,000	49.26	5.01	125.3%	0.0137
8	1	1	1	8	125,000	225.95	1.09	13.7%	0.0628
8	1	2	2	4	125,000	73.93	3.34	41.7%	0.0205
8	1	4	4	2	125,000	43.36	5.69	71.2%	0.0120
8	1	8	8	1	125,000	60.17	4.10	51.3%	0.0167
16	1	1	1	16	62,500	215.40	1.15	7.2%	0.0598
16	1	2	2	8	62,500	70.30	3.51	21.9%	0.0195
16	1	4	4	4	62,500	30.56	8.08	50.5%	0.0085
16	1	8	8	2	62,500	35.64	6.93	43.3%	0.0099
16	1	16	16	1	62,500	85.47	2.89	18.1%	0.0237
32	1	1	1	32	31,250	216.50	1.14	3.6%	0.0601
32	1	2	2	16	31,250	58.24	4.24	13.2%	0.0162
32	1	4	4	8	31,250	25.89	9.53	29.8%	0.0072
32	1	8	8	4	31,250	23.16	10.66	33.3%	0.0064
32	1	16	16	2	31,250	38.65	6.39	20.0%	0.0107
32	1	32	32	1	31,250	29.80	8.28	25.9%	0.0083
64	1	1	1	64	15,625	195.60	1.26	2.0%	0.0543
64	1	2	2	32	15,625	54.11	4.56	7.1%	0.0150
64	1	4	4	16	15,625	20.76	11.89	18.6%	0.0058
64	1	8	8	8	15,625	16.35	15.10	23.6%	0.0045
64	1	16	16	4	15,625	17.23	14.33	22.4%	0.0048
64	1	32	32	2	15,625	9.26	26.66	41.7%	0.0026

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
64	1	64	64	1	15,625	8.14	30.33	47.4%	0.0023
128	1	1	1	128	7,812	208.40	1.18	0.9%	0.0579
128	1	2	2	64	7,812	53.57	4.61	3.6%	0.0149
128	1	4	4	32	7,812	20.45	12.07	9.4%	0.0057
128	1	8	8	16	7,812	14.99	16.47	12.9%	0.0042
128	1	16	16	8	7,812	15.46	15.97	12.5%	0.0043
128	1	32	32	4	7,812	6.40	38.58	30.1%	0.0018
128	1	64	64	2	7,812	4.37	56.46	44.1%	0.0012
128	1	128	128	1	7,812	3.85	64.03	50.0%	0.0011

Table 1b. Runtime scaling for medium-small problem size

scale: medium-small
problem size: 2,240 × 2,240
unknowns: 5,017,600

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
1	1	1	1	1	5,017,600	1,874.00	1.00	100.0%	0.5206
2	1	1	1	2	2,508,800	1,742.00	1.08	53.8%	0.4839
2	1	2	2	1	2,508,800	1,136.00	1.65	82.5%	0.3156
4	1	1	1	4	1,254,400	1,747.00	1.07	26.8%	0.4853
4	1	2	2	2	1,254,400	1,135.00	1.65	41.3%	0.3153
4	1	4	4	1	1,254,400	1,052.00	1.78	44.5%	0.2922
8	1	1	1	8	627,200	1,767.00	1.06	13.3%	0.4908
8	1	2	2	4	627,200	1,148.00	1.63	20.4%	0.3189
8	1	4	4	2	627,200	1,053.00	1.78	22.2%	0.2925
8	1	8	8	1	627,200	1,067.00	1.76	22.0%	0.2964
16	1	2	2	8	313,600	1,084.00	1.73	10.8%	0.3011
16	1	4	4	4	313,600	828.80	2.26	14.1%	0.2302
16	1	8	8	2	313,600	817.30	2.29	14.3%	0.2270
16	1	16	16	1	313,600	810.10	2.31	14.5%	0.2250
32	1	4	4	8	156,800	531.90	3.52	11.0%	0.1477
32	1	8	8	4	156,800	365.30	5.13	16.0%	0.1015
32	1	16	16	2	156,800	332.40	5.64	17.6%	0.0923
32	1	32	32	1	156,800	330.30	5.67	17.7%	0.0917
64	1	8	8	8	78,400	242.70	7.72	12.1%	0.0674
64	1	16	16	4	78,400	156.30	11.99	18.7%	0.0434
64	1	32	32	2	78,400	135.40	13.84	21.6%	0.0376
64	1	64	64	1	78,400	135.30	13.85	21.6%	0.0376
128	1	16	16	8	39,200	118.90	15.76	12.3%	0.0330
128	1	32	32	4	39,200	76.01	24.65	19.3%	0.0211
128	1	64	64	2	39,200	64.76	28.94	22.6%	0.0180
128	1	128	128	1	39,200	64.02	29.27	22.9%	0.0178

Table 1c. Runtime scaling for medium problem size

scale: medium

problem size: 3,160 × 3,160

unknowns: 9,985,600

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
1	1	1	1	1	9,985,600	3,709.50	1.00	100.0%	1.0304
2	1	1	1	2	4,992,800	3,462.00	1.07	53.6%	0.9617
2	1	2	2	1	4,992,800	2,237.00	1.66	82.9%	0.6214
4	1	1	1	4	2,496,400	3,477.00	1.07	26.7%	0.9658
4	1	2	2	2	2,496,400	2,243.50	1.65	41.3%	0.6232
4	1	4	4	1	2,496,400	2,064.00	1.80	44.9%	0.5733
8	1	1	1	8	1,248,200	3,466.00	1.07	13.4%	0.9628
8	1	2	2	4	1,248,200	2,275.00	1.63	20.4%	0.6319
8	1	4	4	2	1,248,200	2,068.00	1.79	22.4%	0.5744
8	1	8	8	1	1,248,200	2,145.00	1.73	21.6%	0.5958
16	1	2	2	8	624,100	2,155.00	1.72	10.8%	0.5986
16	1	4	4	4	624,100	1,763.00	2.10	13.2%	0.4897
16	1	8	8	2	624,100	1,797.00	2.06	12.9%	0.4992
16	1	16	16	1	624,100	1,838.00	2.02	12.6%	0.5106
32	1	4	4	8	312,050	1,090.00	3.40	10.6%	0.3028
32	1	8	8	4	312,050	824.10	4.50	14.1%	0.2289
32	1	16	16	2	312,050	809.70	4.58	14.3%	0.2249
32	1	32	32	1	312,050	804.00	4.61	14.4%	0.2233
64	1	8	8	8	156,025	530.80	6.99	10.9%	0.1474
64	1	16	16	4	156,025	366.20	10.13	15.8%	0.1017
64	1	32	32	2	156,025	331.90	11.18	17.5%	0.0922
64	1	64	64	1	156,025	329.70	11.25	17.6%	0.0916
128	1	16	16	8	78,012	243.10	15.26	11.9%	0.0675
128	1	32	32	4	78,012	157.40	23.57	18.4%	0.0437
128	1	64	64	2	78,012	136.70	27.14	21.2%	0.0380
128	1	128	128	1	78,012	136.50	27.18	21.2%	0.0379

Table 1d. Runtime scaling for large problem size

scale: large
problem size: 4,500 × 4,500
unknowns: 20,250,000

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
1	1	1	1	1	20,250,000	7,546.00	1.00	100.0%	2.0961
2	1	1	1	2	10,125,000	7,031.00	1.07	53.7%	1.9531
2	1	2	2	1	10,125,000	4,520.50	1.67	83.5%	1.2557
4	1	1	1	4	5,062,500	7,088.00	1.06	26.6%	1.9689
4	1	2	2	2	5,062,500	4,569.50	1.65	41.3%	1.2693
4	1	4	4	1	5,062,500	4,191.50	1.80	45.0%	1.1643
8	1	1	1	8	2,531,250	7,149.00	1.06	13.2%	1.9858
8	1	2	2	4	2,531,250	4,474.50	1.69	21.1%	1.2429
8	1	4	4	2	2,531,250	3,876.00	1.95	24.3%	1.0767
8	1	8	8	1	2,531,250	4,289.00	1.76	22.0%	1.1914
16	1	2	2	8	1,265,620	4,390.00	1.72	10.7%	1.2194
16	1	4	4	4	1,265,620	2,313.00	3.26	20.4%	0.6425
16	1	8	8	2	1,265,620	3,624.00	2.08	13.0%	1.0067
16	1	16	16	1	1,265,620	3,767.50	2.00	12.5%	1.0465
32	1	4	4	8	632,812	2,214.50	3.41	10.6%	0.6151
32	1	8	8	4	632,812	1,793.50	4.21	13.1%	0.4982
32	1	16	16	2	632,812	1,826.50	4.13	12.9%	0.5074
32	1	32	32	1	632,812	1,868.50	4.04	12.6%	0.5190
64	1	8	8	8	316,406	1,116.00	6.76	10.6%	0.3100
64	1	16	16	4	316,406	847.05	8.91	13.9%	0.2353
64	1	32	32	2	316,406	830.50	9.09	14.2%	0.2307
64	1	64	64	1	316,406	826.20	9.13	14.3%	0.2295
128	1	16	16	8	158,203	547.10	13.79	10.8%	0.1520
128	1	32	32	4	158,203	376.30	20.05	15.7%	0.1045
128	1	64	64	2	158,203	341.85	22.07	17.2%	0.0950
128	1	128	128	1	158,203	340.45	22.16	17.3%	0.0946

Table 1e. Runtime scaling for very-large problem size

scale: very-large

problem size: 6,400 × 6,400

unknowns: 40,960,000

total cores	nodes	ppn	MPI ranks	threads	unknowns/core	runtime mean/s	speedup	efficiency	CUs
1	1	1	1	1	40,960,000	15,415.00	1.00	100.0%	4.2819
2	1	1	1	2	20,480,000	14,170.00	1.09	54.4%	3.9361
2	1	2	2	1	20,480,000	9,139.00	1.69	84.3%	2.5386
4	1	1	1	4	10,240,000	14,265.00	1.08	27.0%	3.9625
4	1	2	2	2	10,240,000	9,245.50	1.67	41.7%	2.5682
4	1	4	4	1	10,240,000	8,464.00	1.82	45.5%	2.3511
8	1	1	1	8	5,120,000	14,125.00	1.09	13.6%	3.9236
8	1	2	2	4	5,120,000	9,360.00	1.65	20.6%	2.6000
8	1	4	4	2	5,120,000	8,503.00	1.81	22.7%	2.3619
8	1	8	8	1	5,120,000	8,702.00	1.77	22.1%	2.4172
16	1	2	2	8	2,560,000	8,845.00	1.74	10.9%	2.4569
16	1	4	4	4	2,560,000	7,311.00	2.11	13.2%	2.0308
16	1	8	8	2	2,560,000	7,375.00	2.09	13.1%	2.0486
16	1	16	16	1	2,560,000	7,558.00	2.04	12.7%	2.0994
32	1	4	4	8	1,280,000	4,484.00	3.44	10.7%	1.2456
32	1	8	8	4	1,280,000	3,651.00	4.22	13.2%	1.0142
32	1	16	16	2	1,280,000	3,685.00	4.18	13.1%	1.0236
32	1	32	32	1	1,280,000	3,825.00	4.03	12.6%	1.0625
64	1	8	8	8	640,000	2,271.00	6.79	10.6%	0.6308
64	1	16	16	4	640,000	1,826.00	8.44	13.2%	0.5072
64	1	32	32	2	640,000	1,859.00	8.29	13.0%	0.5164
64	1	64	64	1	640,000	1,905.00	8.09	12.6%	0.5292
128	1	16	16	8	320,000	1,140.00	13.52	10.6%	0.3167
128	1	32	32	4	320,000	866.40	17.79	13.9%	0.2407
128	1	64	64	2	320,000	850.80	18.12	14.2%	0.2363
128	1	128	128	1	320,000	846.90	18.20	14.2%	0.2352